

HUECTECH

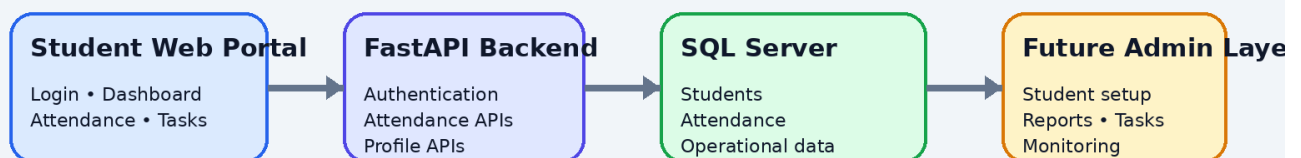
Internship Portal

Client & Employee Project Explanation

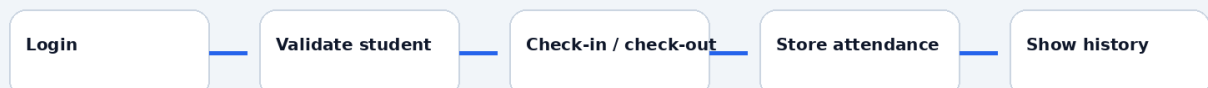
Executive purpose

The HUECTECH Internship Portal is a web-based prototype designed to give interns a single place to authenticate, view their profile, mark daily attendance, access learning tasks and review attendance history.

HUECTECH Internship Portal — Solution Architecture



Core flow



1. Problem Statement

- Intern attendance and learning activity can become difficult to manage when identity, attendance and task information are handled separately.
- The portal addresses this by creating a focused student experience with a backend API and centralized SQL Server data store.

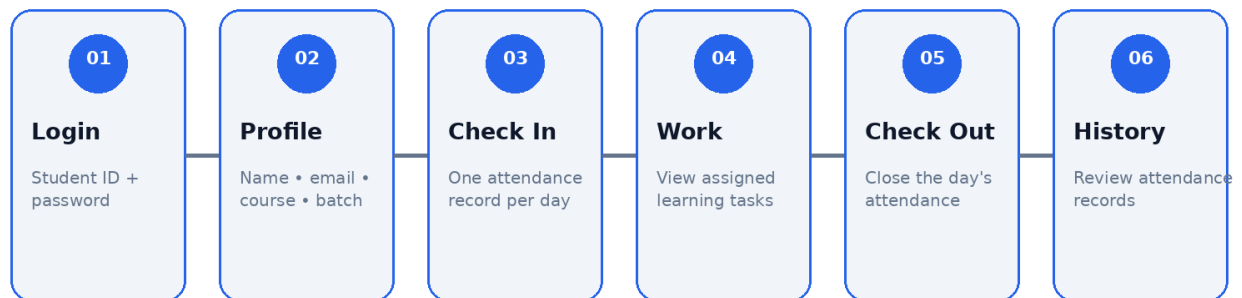
2. Project Initiation

- Identify the intern journey and administrative pain points.
- Define minimum viable requirements: login, student profile, daily attendance and attendance history.
- Design the workflow before implementation.
- Select a lightweight web stack: HTML/CSS/JavaScript + FastAPI + SQL Server.
- Implement modules incrementally and validate each API/UI interaction.
- Prepare the solution for future administration, reporting and task-management features.

3. Solution Flow

- Student opens portal → authenticates with Student ID and password → receives student context → dashboard loads profile and attendance → student checks in → backend validates daily uniqueness → record is stored → dashboard reflects status → student checks out → history remains available.

Student Experience — End-to-End Journey



Business outcome: a single digital touchpoint for interns to authenticate, record attendance and follow daily le

4. Technology Stack

- Frontend: HTML5, CSS3 and vanilla JavaScript.
- Backend: Python FastAPI with Pydantic models.
- Database: Microsoft SQL Server accessed through pyodbc.
- Security: bcrypt password hashing/verification and environment-based database configuration.

5. Functional Modules

- Authentication: StudentCode + password verification.
- Profile: name, StudentCode, email, course, batch and active state.
- Attendance: check-in, check-out, today's status and history.
- Tasks: current UI contains Python, SQL and ADF learning-task examples.
- UI: branded login/dashboard experience with responsive CSS.

Intern Dashboard



Test Student

Student ID: HUEC20260001 • Azure Data Engineering • AUG-2026

Today's Attendance

PRESENT

Check In

09:15 AM

Check Out

06:30 PM

My Tasks

Python Assignment

Pending

SQL Exercise

Pending

Build ADF Pipeline

Pending

Attendance History

Date	Check In	Check Out	Status
2026-09-18	09:15 AM	06:30 PM	Present

6. Attendance Rules

- Only active students can perform attendance operations.
- Duplicate check-in for the same student/date is rejected.
- Check-out requires a prior check-in.
- Duplicate check-out is rejected.
- Attendance history is returned newest-first.

7. Client Value

- Centralizes the daily intern workflow.
- Improves visibility for students and administrators.
- Creates consistent attendance rules.
- Provides an API foundation for future modules.
- Creates a professional digital touchpoint for the HUECTECH internship program.

8. Client Demo Script

- Open the login page.
- Use a controlled test student.
- Show the student profile.
- Perform check-in and explain the validation rule.
- Show the current attendance status.
- Perform check-out.
- Open attendance history.
- Explain the architecture and future roadmap.

9. Production Readiness Notes

- The supplied source package is a prototype and needs a short hardening phase before production.
- main.py currently imports the router from auth.py while the POST /login route is defined in auth_routes.py; the correct router registration should be verified.
- profile.py defines the profile router, but main.py does not currently include it; register it before relying on the profile endpoint.
- The frontend API URL is hard-coded to localhost and should be externalized for deployment.
- Production CORS, authentication/session handling, access control, audit logging, backups and monitoring should be implemented.

10. Roadmap

- Phase 1: stabilize authentication/profile routing and deployment configuration.
- Phase 2: admin dashboard and student management.
- Phase 3: persistent task management and completion tracking.
- Phase 4: attendance calendar, reports and analytics.
- Phase 5: notifications, role-based access and production monitoring.